

CODINGDIDI SERIES

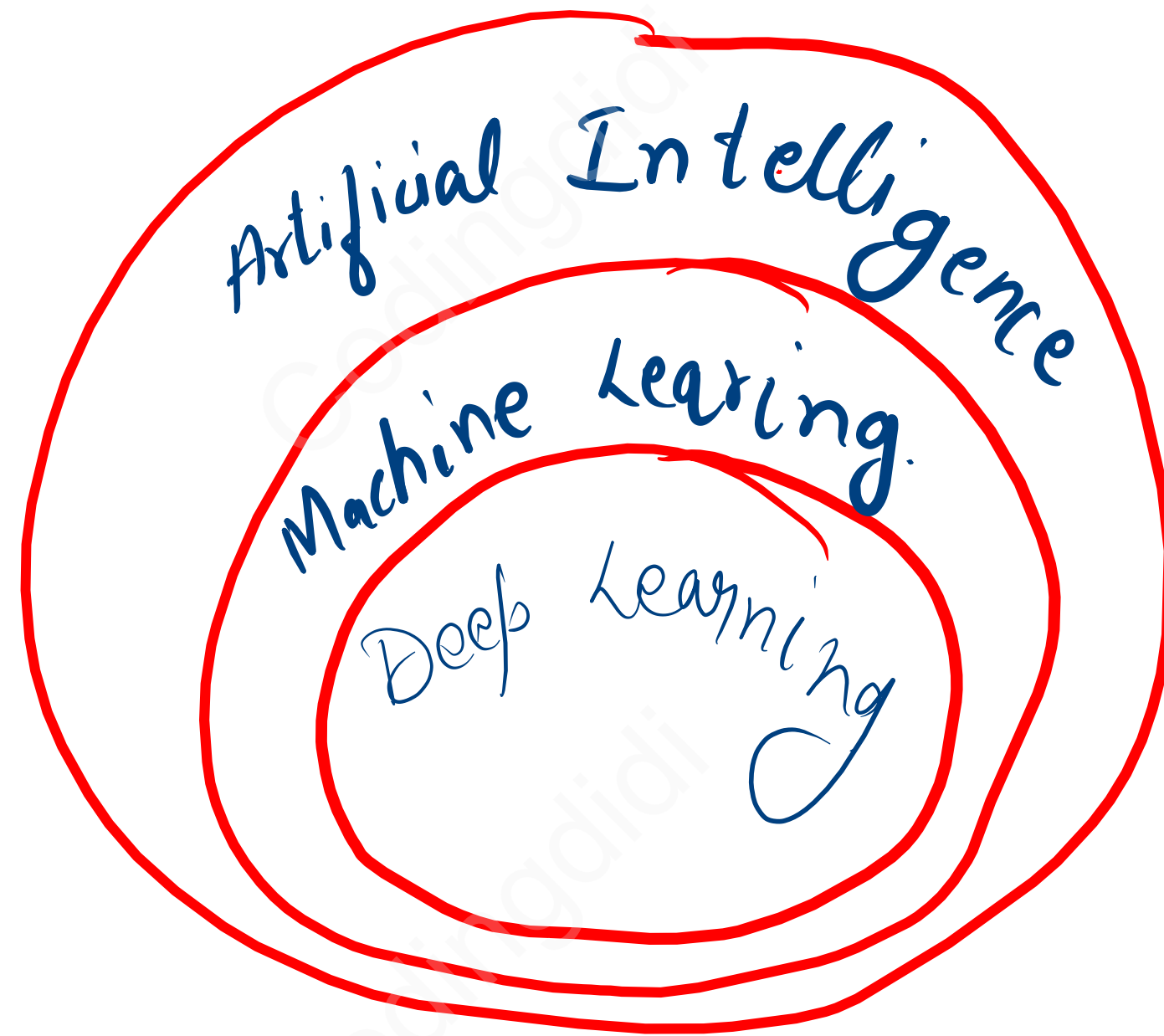
MACHINE LEARNING

FROM SCRATCH

USING  PYTHON



What is Machine Learning?

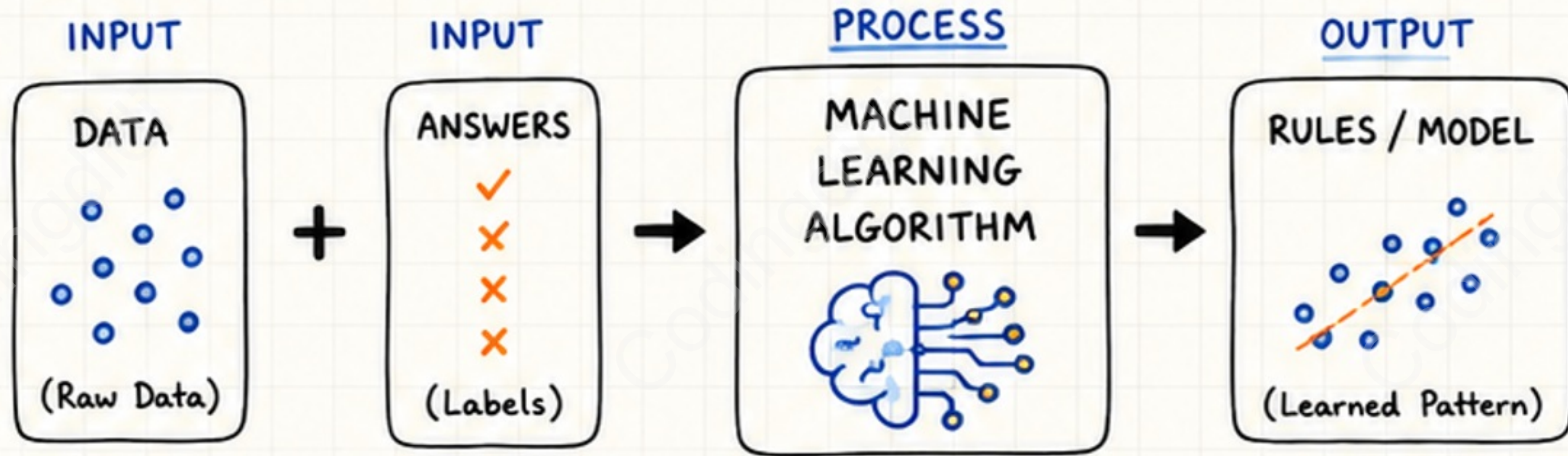


① WHAT IS ML?

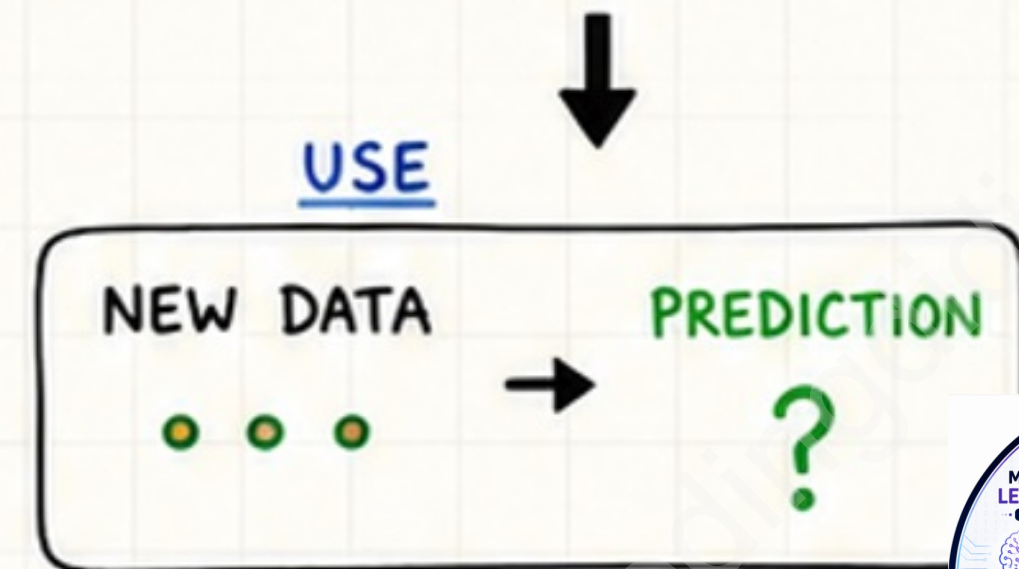
Machine Learning is a way to teach computers to learn patterns from data and make predictions without being explicitly programmed.



② HOW DOES ML WORK?



ML learns the relationship between data (x) and answers (y) to make accurate predictions on new data.



$$\hat{y} = f(x; \theta)$$

We choose θ to minimize error (Loss):

$$J(\theta) = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2$$



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$\hat{y}$ = predicted Value

X = Input features

θ = Model parameters

y = Actual Value.

$J(\theta)$ = Loss / Error

n = No. of training examples.



$$\hat{y} = f(x; \theta)$$

↑ ↑ ↓
predicted input parameters
value features (weights)

We choose θ to minimize error (Loss):

$$J(\theta) = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2$$



Types of ML Model

Supervised
Learning

Unsupervised
Learning

Reinforcement
Learning.



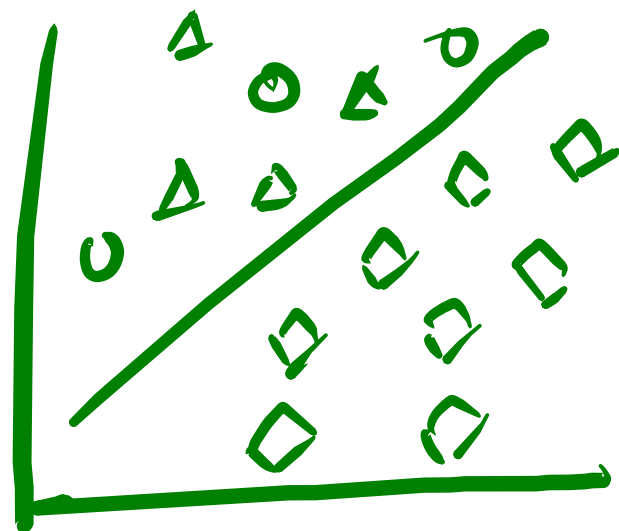
Supervised Learning $\Rightarrow$ Data has
answers/Labels



Supervised Learning $\Rightarrow$ Data has answers/Labels.



Goal $\rightarrow$ predict the output

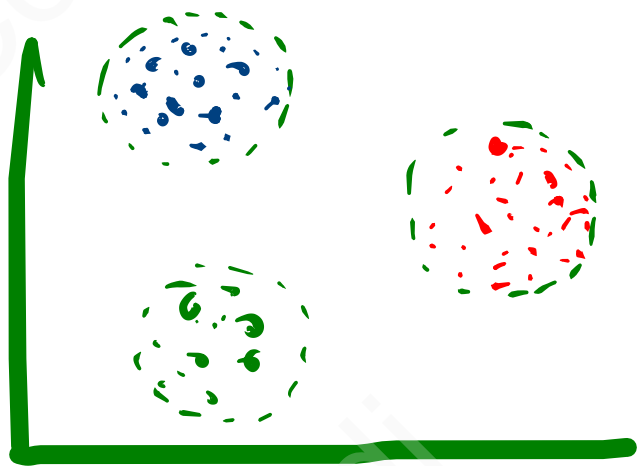


Unsupervised Learning $\Rightarrow$ Data has
no answers/Labels.



Unsupervised Learning $\Rightarrow$ Data has
no answers/Labels.

Goal $\rightarrow$ find patterns/
structures.



Reinforcement Learning $\Rightarrow$ Agent learn.
by $T \& E$ (Trial &
Error)



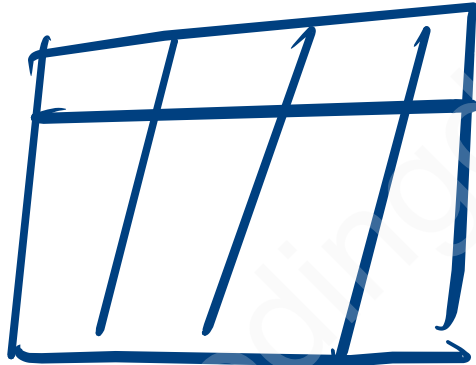
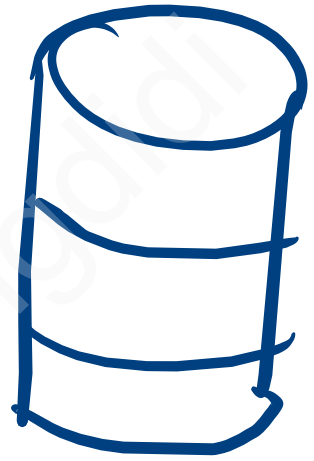
Reinforcement Learning $\Rightarrow$ Agent learn.

by T & ϵ (Trial &
Error)

* Reward
or
* Feedback

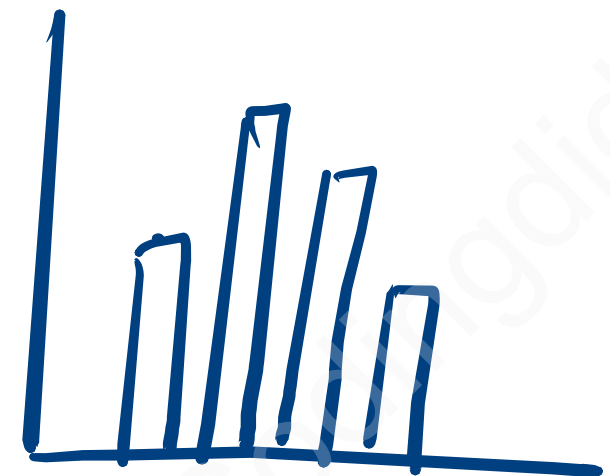
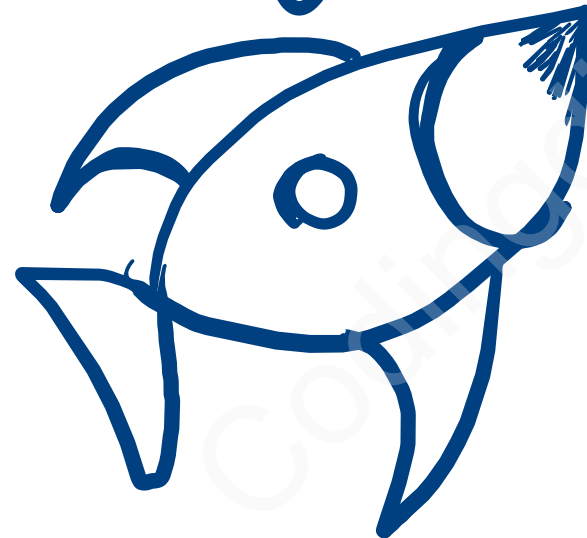


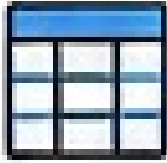
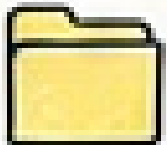
① Collect Data ② prepare Data ③ Train Model.



⑤ Deploy & Predict

④ Evaluate Model



LIBRARY	PURPOSE	BEST FOR
 NumPy	Numerical computing, arrays	Math operations
 Pandas	Data loading & preprocessing	Data cleaning
 Scikit-learn	ML algorithms	Beginner to Intermediate
 SciPy	Scientific computing	Optimization & statistics
 Joblib	Save / load models	Model deployment

